

Combinational Sequential Logic

2.2.1 Combinational and sequential logic design

2.2.1 Combinational and Sequential Logic Design (Detailed Explanation)

English + Nepali Mixup (No Romanize)

Introduction

Digital electronics मा logic circuits लाई मुख्य रूपमा 2 प्रकार मा divide गरिन्छ।

1. **Combinational Logic Design**
2. **Sequential Logic Design**

यी दुवै circuits computer organization, DSP hardware, microprocessor, communication systems, embedded systems लगायत धेरै digital devices मा प्रयोग हुन्छन्।

Difference बुझ्न सजिलो तरीका:

- **Combinational Logic = Present Input मात्र हेर्छ**
- **Sequential Logic = Present Input + Previous State (Memory) हेर्छ**

1. Combinational Logic Design

Definition

Combinational logic circuit भनेको यस्तो digital circuit हो जसको **output केवल current input values मा मात्र depend हुन्छ।**

यसमा कुनै memory हुँदैन।

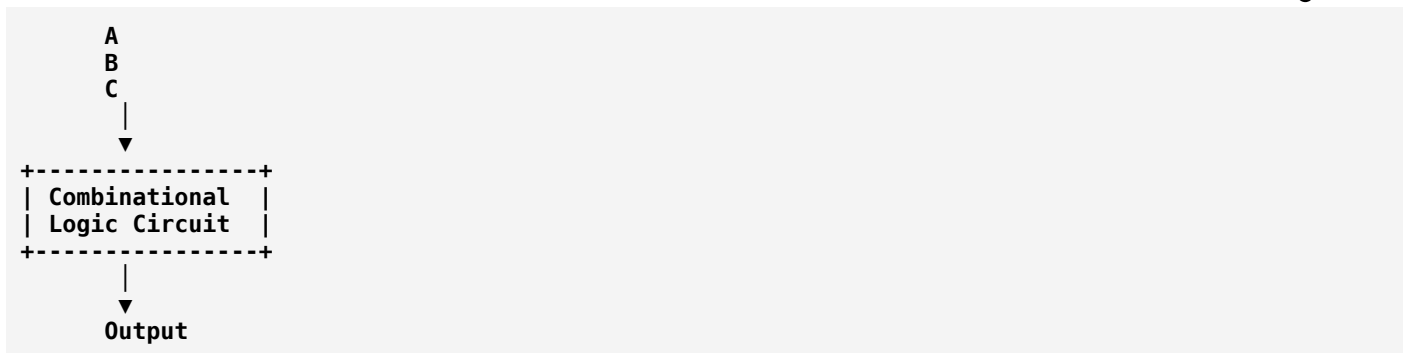
यदि input change भयो भने output पनि तुरुन्त change हुन्छ।

Simple Formula

Output = Function(Current Inputs)

Memory = No

Basic Block Diagram



Output केवल input बाट calculate हुन्छ।

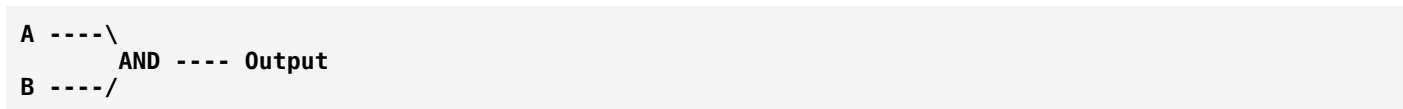
Characteristics

- Memory हुँदैन
- Clock आवश्यक पर्दैन
- Previous output याद राख्दैन
- Faster operation
- Design सरल हुन्छ
- Boolean Algebra मा आधारित हुन्छ

Common Components

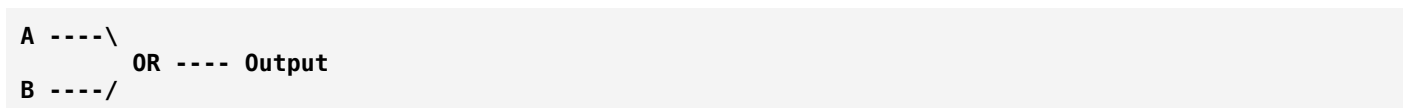
Combinational circuits विभिन्न logic gates प्रयोग गरेर बनाइन्छ।

AND Gate



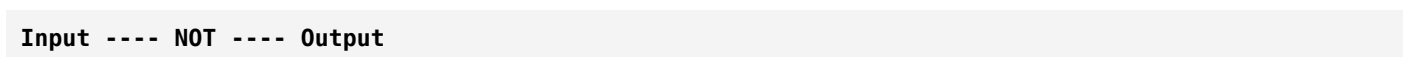
Output = 1 only if both inputs are 1.

OR Gate



Any input 1 भए output 1 हुन्छ।

NOT Gate



Input उल्टो बनाउँछ।

NAND Gate

Universal Gate

AND + NOT

NOR Gate

Universal Gate

OR + NOT

XOR Gate

Different inputs हुँदा Output = 1

Example

0 XOR 1 = 1

1 XOR 0 = 1

1 XOR 1 = 0

Examples of Combinational Circuits

1. Half Adder

Function

Two binary numbers add गर्छ।

Inputs

A
B

Outputs

Sum
Carry

Truth Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

2. Full Addder

Three inputs हुन्छ

A
B
Carry In

Outputs

Sum
Carry Out

3. Multiplexer (MUX)

Many Inputs

↓

One Output

Example

4 Inputs
↓
1 Output

Data select गर्न प्रयोग हुन्छ।

4. Demultiplexer (DEMUX)

One Input

↓

Many Outputs

5. Encoder

Many Inputs

↓

Binary Output

6. Decoder

Binary Input



Many Outputs

7. Comparator

Two binary numbers compare गर्छ।

Output

Equal

Greater

Smaller

Real Life Example of Combinational Logic

Calculator

5 + 3



Result = 8

Calculator ले previous calculation याद गर्दैन।

हर input independently calculate हुन्छ।

Advantages

- Fast
- Easy to design
- Low delay
- No clock needed
- Less hardware

Disadvantages

- Memory हुँदैन
- Previous data store गर्न सक्दैन
- Complex operations गर्न सक्दैन

2. Sequential Logic Design

Definition

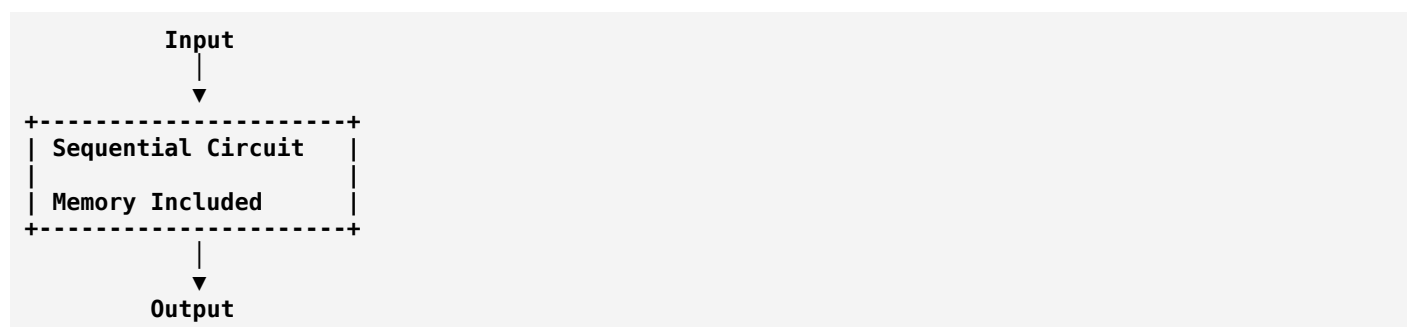
Sequential logic circuit भनेको यस्तो circuit हो जसको output **current input** मात्र होइन **previous state** मा पनि **depend हुन्छ**।

यसमा memory हुन्छ।

Formula

$$\text{Output} = \text{Current Input} + \text{Previous State}$$

Block Diagram



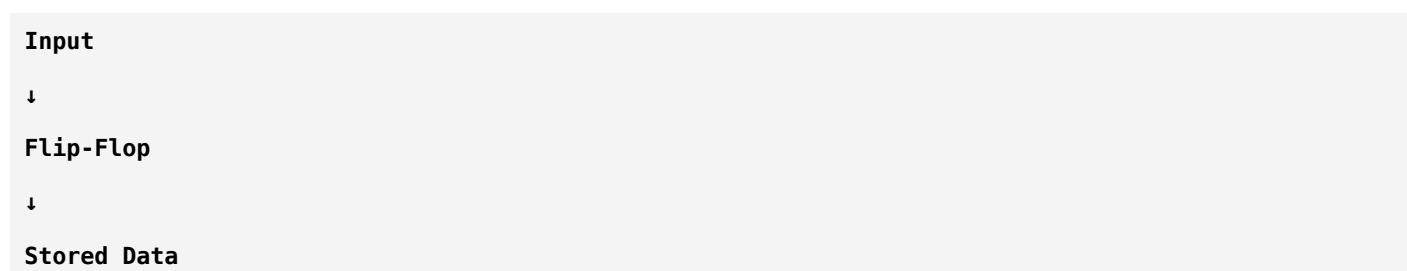
Characteristics

- Memory हुन्छ
- Previous state remember गर्छ
- Clock signal आवश्यक हुन्छ
- Complex operations गर्न सक्छ
- State change हुन्छ

Memory Element

Sequential circuit मा memory राख्ने element

Flip-Flop



Latch

Small memory device

Register

Many flip-flops मिलेर register बन्छ।

Example

```
8 Flip-Flops
↓
8-bit Register
```

Counter

Clock count गर्छ।

Example

```
0
↓
1
↓
2
↓
3
↓
4
```

Types of Sequential Logic

1. Synchronous Sequential Circuit

Clock प्रयोग गर्छ।

Example

```
Clock
↓
All flip-flops update together
```

Advantages

- Accurate
- Stable
- Easy debugging

2. Asynchronous Sequential Circuit

Clock प्रयोग गर्दैन।

Input change भएपछि तुरुन्त output change हुन्छ।

Advantages

Very Fast

Disadvantages

Design difficult

Flip-Flop Types

SR Flip-Flop

Set

Reset

JK Flip-Flop

Improved SR Flip-Flop

D Flip-Flop

Stores one bit.

T Flip-Flop

Toggle operation गर्छ।

Applications of Sequential Logic

- Digital Clock
- CPU Registers
- RAM
- Traffic Light Controller
- Washing Machine Controller
- ATM Machine

- Elevator Control
- Mobile Phones
- DSP Processors
- Communication Systems

Real Life Example

Traffic Light

```

Red
↓
Green
↓
Yellow
↓
Red
  
```

Current light depends on previous light.

Memory आवश्यक हुन्छ।

Password Lock

```

Enter Password
↓
System remembers previous digits
↓
Correct Password
↓
Door Opens
  
```

Memory बिना काम गर्दैन।

Difference Between Combinational and Sequential Logic

Feature	Combinational Logic	Sequential Logic
Memory	No	Yes
Depends On	Current Input	Current Input + Previous State
Clock	Not Required	Usually Required

Feature	Combinational Logic	Sequential Logic
Speed	Faster	Slightly Slower
Complexity	Simple	Complex
Feedback	No	Yes
State	No State	Has State
Storage	Not Possible	Possible
Examples	Adder, MUX, Decoder	Register, Counter, Flip-Flop

Comparison Diagram



DSP Hardware Perspective

DSP processor मा **Combinational Logic** arithmetic operations (AND, OR, Adder, Multiplexer, ALU आदि) को लागि प्रयोग हुन्छ, जबकि **Sequential Logic** registers, counters, state machines, instruction execution, pipeline stages, program counter, र data storage का लागि प्रयोग हुन्छ। आधुनिक DSP hardware ले दुवै प्रकारका logic circuits लाई combine गरेर high-speed signal processing, filtering, FFT, audio processing, image processing, र real-time communication tasks efficiently execute गर्छ।

Key Exam Points

- **Combinational Logic:** Output depends **only on current inputs, no memory, no clock.**

- **Sequential Logic:** Output depends on **current inputs + previous state**, has memory, usually **uses a clock**.
- Common combinational circuits: **Adder, Subtractor, Multiplexer, Demultiplexer, Encoder, Decoder, Comparator**.
- Common sequential circuits: **Latch, Flip-Flop, Register, Counter, Finite State Machine (FSM)**.
- Sequential circuits are essential for **data storage, timing, synchronization, and control** in digital systems.

image illustration for this

